

The empirical recurrence for OEIS A317732, recovered from its tail

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Abstract

OEIS A317732 records “Empirical recurrence of order 81” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 114$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 82$ — every index at which it can be written down. Its last coefficient is $c_{81} = 52776558133248000 \neq 0$, so no further tail satisfies anything shorter; any order-81 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317732 is “Number of $n \times 7$ 0..1 arrays with every element unequal to 1, 2, 3, 4, 5, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 5824, 554642, 67837952, 8216055128, 990049380944, 119470846769818, 14414412169370696, ...

The entry states:

Empirical recurrence of order 81 (see link above)

As of the “Last modified” line on the live entry (Aug 05 2018 15:50:28, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 114$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 114$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 228$ exact terms from $n = 1$ on, it returns order 81 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{81} c_i a(n-i),$$

c_1	85	c_{22}	−283738038069222079063	c_{43}	18982523316036450790025767	c_{64}
c_2	3489	c_{23}	808113536865774597833	c_{44}	166345744979451544788264355	c_{65}
c_3	93529	c_{24}	−2940442044167278888448	c_{45}	302447308962182461335837125	c_{66}
c_4	560259	c_{25}	6419883921928453896684	c_{46}	7966474010412633769867790	c_{67}
c_5	−714407	c_{26}	−13493459323972428762697	c_{47}	−294210029760895344522895626	c_{68}
c_6	−167992433	c_{27}	44825602309195334943725	c_{48}	−461888821965576759929704322	c_{69}
c_7	−472240552	c_{28}	−76062894359853284631523	c_{49}	−62575244969152976533869439	c_{70}
c_8	−3155059881	c_{29}	111536766480229214408042	c_{50}	341845599979591876001931019	c_{71}
c_9	129585745750	c_{30}	−416405702513464032214057	c_{51}	490060612228379913595362115	c_{72}
c_{10}	−119187384808	c_{31}	540485711598689669896410	c_{52}	96546608466556020867735383	c_{73}
c_{11}	3207634076542	c_{32}	−414806772958656442691429	c_{53}	−269735123970351853394325343	c_{74}
c_{12}	−54734841698682	c_{33}	2679528590259613232788233	c_{54}	−35895146605998598631380227	c_{75}
c_{13}	183686448129465	c_{34}	−2523360000294605200058450	c_{55}	−79398303884611436430184104	c_{76}
c_{14}	−1413804777779827	c_{35}	−385992794097722936540034	c_{56}	148958740975944398084176106	c_{77}
c_{15}	13130137029789840	c_{36}	−12981269881109021377673450	c_{57}	183617685043621318785258744	c_{78}
c_{16}	−51708560041005474	c_{37}	8074239983757877006411160	c_{58}	42558077301419341617120273	c_{79}
c_{17}	270191989284882543	c_{38}	11658129403545779693108889	c_{59}	−55621129487669573416714664	c_{80}
c_{18}	−1661478525102741816	c_{39}	48891640415002826804060655	c_{60}	−63797735644983820841803872	c_{81}
c_{19}	5820827526411166356	c_{40}	−17120704029842272204186119	c_{61}	−14726677457616469546680656	
c_{20}	−22412479167933785414	c_{41}	−59227306542020966801693264	c_{62}	13760245736888138459686400	
c_{21}	101320370084216100478	c_{42}	−141304819083191842280624578	c_{63}	14703905271902600484838400	

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A317732 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{81} - c_1 x^{81-1} - \dots - c_{81}$. Its constant term is $-c_{81} = -52776558133248000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component

of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 81 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 81, so $\deg q = 81$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 81, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 81, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 52776558133248000.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-81 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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